

Sahayak

Sahayak is a low-bandwidth welfare scheme discovery assistant built for NSS Open Projects 2026 - Track 1.1: AI-Based Multilingual Chatbot for Welfare Scheme Awareness.

The prototype helps a user fill a short eligibility profile, discover relevant welfare schemes, ask grounded follow-up questions, and generate a document checklist in English or Hindi.

Live demo: https://parzival1821.github.io/Multilingual_Chatbot/

Demo video: <https://vimeo.com/1200973459?share=copy&fl=sv&fe=ci>

Project Snapshot

- Status: Functional MVP deployed on GitHub Pages.
- Scope: 8 high-impact central welfare schemes.
- Languages: English and Hindi, with Hindi rendered in Devanagari.
- Core AI pattern: Retrieval-grounded answer generation over a curated scheme knowledge base.
- User flow: 4-6 simple eligibility signals produce a personalized scheme shortlist.
- Safety: Unknown questions return a fallback instead of hallucinated scheme details.
- Verification: Local core and static smoke tests pass with npm run check.

NSS Challenge Fit

Official expectation	Current implementation
Focused catalogue of 8-10 schemes	8 schemes included
Eligibility-questioning flow	Guided profile form with area, occupation, gender, income, land, housing, LPG, bank, and more
Personalized shortlist	Ranked recommendations with match reasons
Multilingual interface	English + Hindi Devanagari
Document checklist	Copyable and downloadable scheme-specific checklist
Grounded answers	Retrieval over local scheme data with source links
Low-bandwidth thinking	Static app, text-first UI, no heavy framework, downloadable .txt handoff
Honest unknown handling	Safe fallback for unsupported questions

Scheme Catalogue

Scheme	Primary beneficiary	Current verification
PM-KISAN Samman Nidhi	Landholding farmer families	Verified official source
Ayushman Bharat PM-JAY	Poor and vulnerable families needing health cover	Verified official source
Pradhan Mantri Ujjwala Yojana	Adult women from poor households without LPG	Verified official source
Pradhan Mantri Awas Yojana - Urban 2.0	Urban EWS/LIG/MIG families without a permanent house	Verified official source
National Social Assistance Programme	BPL elderly, widows, disabled persons, and vulnerable households	Verified official source
PM Street Vendor's AtmaNirbhar Nidhi	Urban and peri-urban street vendors	Official source
Atal Pension Yojana	Informal/unorganised workers with bank accounts	Official source
Sukanya Samriddhi Account	Girl child, through parent or guardian	Official source

System Architecture

Diagram: Rendered in the Visual Diagrams PDF.

User Journey

Diagram: Rendered in the Visual Diagrams PDF.

Grounded Answer Flow

Diagram: Rendered in the Visual Diagrams PDF.

Demo Personas

Diagram: Rendered in the Visual Diagrams PDF.

Implementation Details

- index.html and styles.css implement the responsive single-page dashboard.
- src/data.js stores curated scheme data, Hindi names, summaries, labels, documents, benefits, and source links.
- src/core.js contains eligibility scoring, retrieval, profile inference, checklist formatting, and grounded answer composition.
- src/app.js connects the UI to the recommendation engine, assistant, language switcher, and checklist actions.
- tests/core.test.js validates recommendation ranking, retrieval, fallback behavior, Hindi queries, and checklist logic.
- tests/static-smoke.test.js validates the static shell, deployed asset references, Hindi Devanagari copy, and Pages workflow shape.

Submission Documents

The submission/ folder contains Google-Docs-ready Markdown files for the official NSS deliverables:

- [Combined PDF Submission Packet](#)
- [Problem Analysis Report](#)
- [Solution Framework](#)
- [Prototype and Implementation Plan](#)
- [Impact Projection](#)
- [Pilot Test Report](#)
- [Final Presentation Content](#)
- [Submission Portal Details](#)
- [Visual Diagrams](#)

Run Locally

npm run dev

Then open:

<http://localhost:4173>

Test

npm test

Run all local checks:

npm run check

Demo Script

1. Open the live app or run it locally.
2. Click the Farmer preset.
3. Show PM-KISAN as a strong match and PM-JAY as an additional support option.
4. Click View documents on PM-KISAN and show the checklist panel.
5. Copy or download the checklist to demonstrate low-bandwidth handoff.
6. Ask: What documents do I need for Ayushman Bharat?

7. Switch to Hindi and ask: मुझे घर के लिए कौन सी योजना मिल सकती है?
8. Show source links, verification labels, and the safe fallback for unsupported questions.

Verification Checklist

- Farmer persona ranks PM-KISAN first.
- Woman-led household persona shows Ujjwala, PMAY-U, and Sukanya.
- Street vendor persona ranks PM SVANidhi first.
- Hindi mode uses Devanagari labels, prompts, scheme names, and checklist content.
- Recommendation View documents action scrolls to and highlights the checklist panel.
- Checklist copy and download actions produce a scheme-specific document list.
- Unsupported questions produce a safe fallback instead of invented scheme details.
- npm run check passes.

Project Positioning

Sahayak is intentionally built as a practical AI product rather than a custom ML research model. Its strength is the combination of structured eligibility logic, retrieval-grounded answers, bilingual UX, source-backed recommendations, and a low-bandwidth document handoff compatible with WhatsApp, SMS, or IVR channels.